

Multi-Hazard Readiness & Response (pre-event MVP)

Client	Southeastern Grid & Water (SGW) — fictional US utility, 8M+ residents, SC/GA/NC footprint
Engagement	AECOM AI-enabled operational decision-support platform
Audience	SGW technical delivery team
Version	1.1 (condensed)
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Companion documents	01_assumptions.md, 02_mvp_workflow.md, 05_architecture.md, 06_data_model.md, 07_external_data_sources.md

1. Problem definition & business context

SGW's operational data is fragmented across four system families procured independently over decades — **GIS** (asset registry, hazard overlays; ESRI-shaped), **CMMS** (work orders, inspections; Maximo-shaped), **SCADA/IoT** (real-time telemetry) and **field operations** (crews, dispatch, reports). Each uses its own asset identifier and refresh cadence; external NOAA data (NWS, NHC, NOS CO-OPS, Digital Coast, SPC/CPC, NCEI) is consumed piecemeal. During a severe-weather event an operator consults four to seven screens to understand what is happening to which asset, with which crews, under which forecast — so the platform of record for "what's happening now" is the operator's short-term memory.

Three pressures make this a now-problem: climate-driven event frequency across the SC/GA/NC footprint (hurricane, coastal + inland flood, heatwave, wildfire — see A5), insurer and PUC demands for demonstrable resilience posture under tightening NERC CIP obligations, and rising operational cost — unplanned outages, storm overtime and reactive maintenance are the largest controllable cost lines.

What this platform is: an AI-enabled decision-support platform that unifies fragmented data into a single situational picture, applies the right AI technique to each sub-problem (§5), and keeps operators in control — every recommendation advisory, every action auditable. **What it is not:** a chatbot, a resident-facing product, or a replacement for operator judgement.

2. Key assumptions & unknowns

Full register in *01_assumptions.md*. The assumptions most load-bearing for technical scoping:

#	Assumption	If wrong, this changes
A1	Personas are NOC / Emergency / Field / Maintenance — internal operational users, not the 8M residents.	Persona-specific workflows re-prioritise; a resident-facing product is fundamentally different scope.
A3	SGW reference footprint is SC/GA/NC (coastal + inland).	Different footprint uses the same adapters with region-specific hazard fixtures; workflow unchanged.
A4	NOAA is the reference federal data stack; adapter pattern isolates provider.	Non-US (Met Office, ECMWF, JMA) requires adapter implementations only; risk/optimisation/UI unchanged.
C1	Hybrid deployment — cloud for training/batch, on-prem edge appliance at NOC for real-time inference and continuity during outages.	Cloud-only fails FERC/NERC CIP and utility resilience credibility. Cloud-native with hardened redundant WAN simplifies.

#	Assumption	If wrong, this changes
D1	All AI recommendations are advisory; a human in the appropriate role must accept, override or comment before action.	Regulatory posture and adoption depend on this; not negotiable for MVP.
E1	MVP is the pre-event / early-onset phase for the four named hazards, coastal + inland.	Storm-only scope under-serves inland exposure; during-/post-event are Phase 2/3.
E3	Mock dataset is fragmented on purpose (multi-format, ID crosswalk, quality flags, freshness metadata).	If SGW's data is already unified, the ingestion workstream shrinks — AI portfolio and workflow unchanged.

Unknowns that most affect scoping — with 15 minutes of an SGW operations director we would prioritise: (1) the single biggest pain during a major storm — *knowing*, *deciding*, or *executing*? (shifts the MVP phase); (2) which decision-support tools operators already trust (anchors change-management); (3) where SGW's data is most and least mature (reshapes the roadmap).

3. Target users, workflows & pain points

Persona	Today's pain	What this platform gives them
NOC Operations Controller	Disconnected dashboards; no single view of asset health under weather stress	Live map with hazard-conditional risk overlay + ranked flagged-asset list + drill-down with explanation
Emergency Response Coordinator	No single operational picture; decisions rely on memory and phone calls	Shared operational view + cascading-impact chains + on-demand exec briefing draft
Field Operations Supervisor	Constant re-prioritisation; suboptimal crew placement	Optimised crew pre-positioning with trade-off transparency + accept/override workflow
Maintenance Planner	Reactive work dominates; preventative work slips when it matters most	Preventative work-order recommendations sequenced by risk × consequence

Secondary (view-only in MVP): Regional Operations Managers, Executives (Deliverable 2 lives here), Data/GIS stewards. Residents, regulators and insurers are stakeholders, not users.

Anchor workflow — pre-event / early-onset (full detail in *02_mvp_workflow.md*): a weather alert or SCADA anomaly within the 72-hour horizon triggers ingestion and joins to GIS + hazard zones; hazard-conditional risk scoring ranks at-risk assets; Prophet forecasts stress, water levels and demand with uncertainty bands; dependency-graph traversal surfaces cascading impacts (substation → pumping station → hospital) and Louvain clustering groups blast-radius; the optimiser proposes crew pre-positioning; the LLM synthesises explanations citing evidence IDs; the operator accepts / overrides / comments with everything audit-logged; an executive briefing is drafted on demand.

4. Functional & non-functional requirements

Functional requirements

Ingestion (FR-1-7) — ingest NWS alerts/forecasts/observations at configurable cadence; NHC cones + SLOSH layers; NOS CO-OPS water levels (anchor: Charleston Harbor 8665530); Digital Coast flood layers pre-clipped to SC/GA/NC and reprojected to WGS84; GIS registry, CMMS work orders, SCADA telemetry (quality flags preserved) and field reports. Source-specific asset IDs resolve to a canonical `asset_id` via the crosswalk; per-record freshness metadata flows through the pipeline.

AI & analytics (FR-8-14) — hazard-conditional risk scoring per asset with contributing factors and confidence; Prophet forecasting with weather as exogenous regressor and uncertainty bands; Prophet-residual anomaly detection on SCADA; dependency-graph traversal (networkx BFS) for cascading impacts; Louvain blast-radius clustering; OR-Tools crew pre-

positioning under coverage, travel-time and shift-hour constraints; structured-output LLM explanations citing evidence IDs from every contributing source.

Operator UI (FR-15-20) — live map with hazard overlay + risk heatmap; ranked at-risk asset list; per-asset drill-down (score, features, dependency chain, evidence, explanation); accept / override / comment on every recommendation; immutable exportable audit log; on-demand executive briefing generation.

Non-functional requirements

Category	Requirement
Availability	99.9% NOC-side during declared event window; edge appliance continues if cloud unreachable
Latency	Dashboard refresh < 5 s; risk refresh < 60 s from new weather data; LLM explanation < 8 s
Data freshness	SCADA < 60 s; weather 1–6 h per product; GIS daily; CMMS 15 min; field reports < 5 min
Auditability	Every recommendation and operator action logged, immutable, exportable machine-readable
Security	NERC CIP alignment; TLS 1.3 in transit, AES-256 at rest; RBAC; MFA; secrets in HSM-backed vault
Accessibility	WCAG 2.1 AA; low-light NOC environments, operation under duress
Observability	Model drift, calibration and fairness metrics; latency SLOs; per-source data-quality alerting
Scalability / portability	Horizontal scaling of ingestion + serving; regional expansion via hazard fixtures + retrained heads; adapter pattern isolates provider for non-US deployment

5. Proposed AI capabilities

Portfolio scoped to techniques with genuine prior applied work; per-capability rationale in *02_mvp_workflow.md*.

#	Capability	Model family	What it does NOT do	Human validation
1	Hazard-conditional risk scoring	LightGBM + Random Forest baseline. Calibration to a true failure probability is Phase 2 (needs historical failure joins)	Emits a raw stress score, not a calibrated probability; does not decide response	Accept/override; monthly hold-out review
2	Time-series forecasting	Prophet + weather exogenous regressors + M2 tidal seasonality; SARIMAX for stationary sub-series	80% band is nominal — empirical coverage ~58% on held-out Debby data (disclosed on Governance)	Uncertainty band surfaced in UI
3	SCADA anomaly detection	Prophet-residual + rolling-median outlier ranking	Does not attribute causes or decide dispatch	Confirm/dismiss feeds recalibration
4	Crew pre-positioning	OR-Tools VRP / Guided Local Search	Does not commit dispatch or set objective weights	Operator adjusts weights, accepts/overrides
5	Cascading impact	networkx BFS over <code>asset_dependencies</code>	Only traverses declared dependencies; no prediction	Stewards maintain graph; operator inspects chain
6	Blast-radius clustering	Louvain community detection	Groups by topology only; no prioritisation	Operator filters by cluster
7	Fairness auditing	Demographic-parity + equal-opportunity gaps by region / domain / demographics	Monitoring signal only; no direct mitigation	Model-risk committee monthly; breaches trigger recalibration

#	Capability	Model family	What it does NOT do	Human validation
8	Copilot explanation layer	Structured-output LLM (gpt - oss : 120b via Ollama Cloud; OpenAI fallback)	Never produces scores, forecasts or plans — narrates and cites only	Drift monitored via canonical eval set
9	Operator-preference alignment	Logistic regression on audit-log accept/override history — deliberately not RL (no reward signal, no exploration on real infra, interpretability required)	Bounded adjustment $ \Delta \leq \beta = 0.15$; cannot flip Critical to Low; weights visible on Governance	Every decision is HITL; force-retrain requires operator action

LLM boundary (non-negotiable): the LLM narrates, cites evidence and drafts; it never produces the risk score, the forecast or the optimisation plan. This split is what makes the platform defensible under model-risk review — decision support, not an ops chatbot.

Phase 2: hazard-onset classifier; GRU-with-exogenous forecasting where Prophet plateaus; Isolation Forest / autoencoder anomaly upgrade. **Phase 3:** computer vision over NGS post-event aerial imagery for damage assessment.

6. High-level architecture & integrations

Full detail in *05_architecture.md*; source registry in *07_external_data_sources.md*.

```
[External data] NOAA (NWS, NHC, NOS CO-OPS, Digital Coast, SPC/CPC, NCEI)
                  |
                  | SGW internal (GIS, CMMS, SCADA/IoT, field ops)
[Ingestion]       |
                  | Six-adapter Hazard Data family + internal adapters
                  | ID resolution (asset_id_crosswalk) · quality flags · freshness · CRS →
WGS84
[Persistence]     |
                  | PostgreSQL 16 + PostGIS 3.4 – GiST spatial indexes; partitioned
                  | sensor_readings; JSONB LLM payloads; append-only audit_log;
                  | operational_risk_snapshot materialised view
[Models]          |
                  | Risk (GBM+RF) | Forecast (Prophet) | Anomaly (residual)
                  | Optimisation (OR-Tools) | Graph (networkx + Louvain)
[Explanation]    |
                  | Structured-output, evidence-citing LLM copilot
[API + UI]        |
                  | FastAPI + React dashboard · hash-chained audit log
[Downstream]      |
                  | CMMS work-order creation | dispatch handoff | exec briefing
```

Adapters: AlertAdapter (NWS alerts), ForecastAdapter (NWS gridpoints; HRRR/GFS Phase 2), ObservationAdapter (NWS stations + CO-OPS gauges), HazardLayerAdapter (Digital Coast, SLOSH, SPC/CPC, FEMA), TrackAdapter (NHC cones), StreamflowAdapter (National Water Model, Phase 2).

Deployment: hybrid — cloud (AWS / Azure GovCloud) for training, batch inference and storage; on-prem edge appliance at the NOC for real-time inference, dashboard and audit write-through, continuing to operate if the cloud is unreachable. Justified by NERC CIP obligations and resilience credibility for a utility whose product is resilience.

Prototype → production: the demo's Docker Compose stack maps 1:1 to the production AWS shape — same Postgres schema on RDS Multi-AZ, same containers on EC2 ASG behind an ALB, same hash-chained audit contract with an S3 Object Lock mirror, same `LLMProvider` adapter with Bedrock (Claude on GovCloud, PrivateLink) substituted by one environment variable, same CI eval gate. The one genuinely new workstream is real ingestion: replacing the mock generator with adapters against SGW's live GIS / CMMS / SCADA, sequenced GIS → CMMS → SCADA → Field Ops behind the same `Fetcher` protocol. Full mapping in *05_architecture.md*.

Integrations: ESRI-shaped GIS (ArcGIS REST + GeoJSON), Maximo-shaped CMMS (REST read/write; work-order creation on operator confirmation), SCADA historian batch + MQTT stream with quality flags, NOAA stack (REST, shapefile/KMZ, ArcGIS REST, anonymous public S3 — note `signature_version=UNSIGNED` for `noaa-nwm-pds` / `noaa-hrrr-bdp-pds`).

7. Data requirements, dependencies & quality

Full spec in `06_data_model.md` and `07_external_data_sources.md`. Eight domains: assets (GIS), maintenance (CMMS), weather + hazard (NOAA), operations (SCADA), field ops, incidents + outages, infrastructure dependencies, and reference (ID crosswalk, regions, historical events).

Fragmented on purpose: the mock dataset reproduces SGW's actual fragmentation — multiple formats, per-source asset IDs requiring crosswalk resolution, realistic sensor quality flags, free-text field notes. A pre-joined dataset would hide the exact integration problem the platform solves; the ingestion + ID-resolution layer delivers value before any AI is invoked.

Risk	Mitigation
Noisy / missing sensor readings	Quality flags preserved end-to-end; anomaly detection distinguishes outlier from sensor fault
Stale weather data	Per-record freshness metadata; staleness surfaced beside every recommendation
Sparse rare-event history (Cat 3+, 100-yr floods)	Physics-informed features (elevation, flood zones, vegetation); transfer from NCEI Storm Events; uncertainty surfaced in UI
Model drift as climate shifts	Continuous calibration monitoring; scheduled retraining; drift alerts
Asset ID mismatches	Crosswalk at ingest; unmapped IDs raise a data-quality alert, never silently dropped
Regional imbalance (coastal better instrumented)	Fairness auditing surfaces gaps; retraining prioritises under-represented regions

Classification & retention: public (NOAA) → internal (registry, work orders, rosters) → sensitive (SCADA, incident detail — NERC CIP exposure) → regulated (audit logs). Operational data and audit logs retained ≥ 7 years; model artefacts versioned indefinitely, retirement requires model-risk sign-off.

8. Security, governance & human oversight

Regulatory frame: NERC CIP (grid-side critical-asset obligations), state PUC reporting, SOC 2 (cloud controls), NIST AI RMF (alignment stated, not certified in MVP).

Human-in-the-loop (non-negotiable): every recommendation is advisory; the appropriate named role confirms before action — crew dispatch (Field Supervisor), asset shutdown (NOC Controller), public communication (Emergency Coordinator), preventative work order (Maintenance Planner). Every accept / override / comment is logged with user, timestamp, reason, and the exact model version + input features that produced the recommendation.

Audit log: immutable, exportable, retained ≥ 7 years — every recommendation (score, model version, features, evidence IDs), every operator action, every retraining event and calibration report, every fairness audit result.

Model risk management: per-region, per-hazard calibration curves with drift alerts; feature- and prediction-distribution drift monitoring; fairness gaps (demographic parity + equal opportunity) across region, utility domain and service-area vulnerability indices — breaches trigger recalibration and reviewer sign-off. Explainability is role-shaped: contributing factors for the Ops Controller, dependency chains for the Emergency Coordinator, dispatch trade-offs for the Field Supervisor, calibration + fairness dashboards for the Model-Risk Reviewer.

Access & LLM governance: RBAC across eight roles, MFA, least-privilege pipelines, secrets in an HSM-backed vault. LLM: structured output only, canonical eval set for drift, prompt versioning with full traceability, data-classification review before any sensitive data reaches a hosted model.

9. Success metrics, MVP scope & delivery priorities

Success metrics — operational, not model-only. Model metrics (AUC, MAE, Brier, forecast coverage) are inputs to the metrics that matter:

Metric	Baseline	Target (12 months post-MVP)
Lead time on preventative work orders during events	<i>TBD from SGW data</i>	+48 h
Avoided outages per major event (customer-hours)	<i>TBD from SGW data</i>	−20%
MTTR during storm events	<i>TBD from SGW data</i>	−15%
Crew utilisation during pre-event window	<i>TBD from SGW data</i>	+25%
Operator time to full operational picture	15–20 min (assumed)	< 3 min
Operator override rate	—	15–25% (too low = over-trust; too high = under-value)

In scope (MVP): pre-event / early-onset phase; four hazards; SC/GA/NC footprint; Hurricane Debby (Aug 2024) demo scenario + Hurricane Idalia (Aug 2023) validation reference; all nine \$5 capabilities (functional, simplified where noted); operator UI with accept/override + audit log; NOAA MVP source stack.

Out of scope (Phase 2/3): live during-event triage; post-event CV damage assessment; cross-domain water optimisation; resident-facing communication; work-order execution tracking; hazard-onset classifier; NWM / HRRR / NGS feeds; non-US adapters (portability designed in, not built).

Delivery priorities (by dependency and value): 1) foundational data engineering — adapters, ID resolution, quality flags (immediate ROI before any AI); 2) risk scoring + baseline; 3) Prophet forecasting; 4) residual anomaly detection; 5) OR-Tools optimisation; 6) graph reasoning; 7) LLM explanation layer; 8) operator UI; 9) governance dashboards; 10) executive briefing view.

Phase 2 highlights (full list in companion docs): NWM streamflow and HRRR gridded forecasts; live during-event triage; real MCP server exposing platform tools to any MCP-compatible agent; full RBAC capability gating per persona; executed decisions wired into Maximo work-order creation and crew dispatch; cross-session agent memory. **Phase 3:** NGS emergency-response imagery with CV damage assessment; NIFC/InciWeb wildfire perimeters.

Appendices: A. Assumptions register (*01_assumptions.md*) · B. Workflow selection (*02_mvp_workflow.md*) · C. Data model (*06_data_model.md*) · D. Source registry (*07_external_data_sources.md*) · E. Architecture (*05_architecture.md*)

Version	Date	Change
1.0	2026-07-15	First complete draft
1.1	2026-07-15	Condensed for submission — detail moved to companion docs